ENAM Holdings — AI Engineer Build Challenge

FINN: Financial Investment Agent · Submission Dossier & Covering Note

Subject: ENAM - IIT screening - 2027	**Candidate Role:** AI Engineer
Submission Email: assessment@enam.com	**Date:** November 2026

LIVE DEPLOYED MOBILE APP (PROTOTYPE):
https://finn-assigbnment-1f4bztefw-vaibhav-guptas-projects-b910ed08.vercel.app/brief

1. Executive Overview & Problem Addressed

Public-markets equity investors are inundated daily by thousands of routine exchange filings, noisy price swings, and generic broker commentary. The scarce skill in public-market investing is not data aggregation; it is **curation and auditability**: separating signal from noise, quantifying management credibility, and presenting actionable primary disclosures in a focused 60-second morning workflow.

FINN was built from the ground up as a mobile-first, high-density financial terminal that rejects 'AI slop' (unformatted LLM hallucinations and purple-gradient templates) in favor of calm editorial finance. It operates cleanly in two modes: **Push** (proactively filtering what changed across the portfolio) and **Pull** (grounded, on-demand primary source research).

2. Flagship Workflows & Differentiating Capabilities

A. Promised vs. Delivered Scorecard (Pull Flagship ■):
Rather than just summarizing earnings calls, FINN extracts management's explicit historical promises (revenue, EBITDA margin, capex, store additions) with verbatim quotes and page citations from the last 4 concall transcripts. It then automatically cross-audits those commitments against subsequent audited financials and quarterly results tables. Gemini 2.5 Pro synthesizes an objective **Executive Credibility Score** (e.g. *8/10 Met*) and highlights exactly where narrative diverged from reality.

B. Materiality-Ranked Daily Brief (Push Mode):
Exchange filings are prioritized by a rigorous mathematical scoring function (*priority* = *category_weight* × *materiality* × *position_weight*). Routine ESOP allotments and trading window notices are collapsed into compact counters. High-impact disclosures expand into custom cards: Quarterly results show 4-row mini tables (Revenue, EBITDA/NII, PBT, PAT) with QoQ and YoY deltas and basis-point margin comments; C-suite management changes pull verified career profiles from SEBI annexures and live search.

C. Threshold Triggers & Correlated 'Why It Moved':
Investors set personalized thresholds for volume multipliers (e.g. >2.0x 20-day trailing avg) and 1-day price swings (±5%). When an unusual move triggers, FINN correlates it against filings from the same 24-48h window to immediately answer *'Why it moved'*.

3. Architectural Cost Efficiency & Model Routing

To maximize intelligence while maintaining near-zero operating costs on serverless tiers, FINN implements tiered model routing with SHA-256 caching:

Pipeline Task	Model / Engine	Architectural Justification	Unit Cost

Filing Classification	Rule-based Taxonomy	Deterministic keyword/BSE mapping filters 85% of filings without LLM calls.	■0.00
Ambiguous Disclosures	Gemini 2.5 Flash	High-speed classification, one-liner summaries (≤18 words), results extraction.	~■0.005
Promised vs Delivered	Gemini 2.5 Pro	Deep reasoning reconciling verbatim concall claims against audited reports.	~■0.75
Vector Retrieval	Gemini Embeddings	Selective chunking of high-signal AR sections (MD&A;, Auditor notes, Segments).	~■0.001

4. Technical Decisions, Trade-Offs & Roadmap

Deliberate Design Decisions:

• **Primary Source Strictness:** Pull research is strictly bounded to Annual Reports, Concall Transcripts, and BSE filings. Broker notes and open-web hallucinations are disallowed by design to ensure 100% auditability.

• **Next.js Serverless + GitHub Actions Cron:** Heavy batch scraping and PyMuPDF PDF chunking run on GitHub Actions cron schedules (13:00 UTC Mon-Fri), keeping the Vercel serverless deployment fast, lightweight, and resilient.

• **Zero-Empty-Screen Offline Fallback:** Every endpoint serves from versioned local snapshots (`data/snapshots/`) when external exchange APIs or databases experience network timeouts.

• **What Was Deliberately Left Out:** Multi-user auth and intraday tick streaming were omitted to preserve focus on high-signal corporate disclosures.

• **Next Roadmap:** Automated morning briefs delivered via WhatsApp, user feedback loops to adjust priority weights based on swipe dismissals, and cross-portfolio capex tracking.

5. Submission Verification & Links

Item	Status & Access Details
Mobile Prototype	https://finn-assigbnment-1f4bztefw-vaibhav-guptas-projects-b910ed08.vercel.app/brief (Optimized for phone & desktop frame)
Architecture & Docs	README.md (architecture diagram, local setup, model routing) + WRITEUP.md
Data Plumbing	18 portfolio stocks + 4 watchlist companies; BSE API + yfinance + Screener.in
AI Build History	Full transcript and step-by-step agent trajectory preserved in repository